

Regresní analýza

LMER model - fixní efekty, K

Mnohem lepší R^2 , stále malé. Nevýznamný poměr S/K a styl využití opce (a model).

```
subset_args <- c(
  #"S",
  "moneyness",
  "K",
  "r",
  "sigma",
  "q",
  "T",
  "am_eu",
  "call_put",
  "model",
  "ticker",
  "snapshot_month"
)
interactions <- c()

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lm(formula, data = train_data_rel)

summary(rmr_relative)

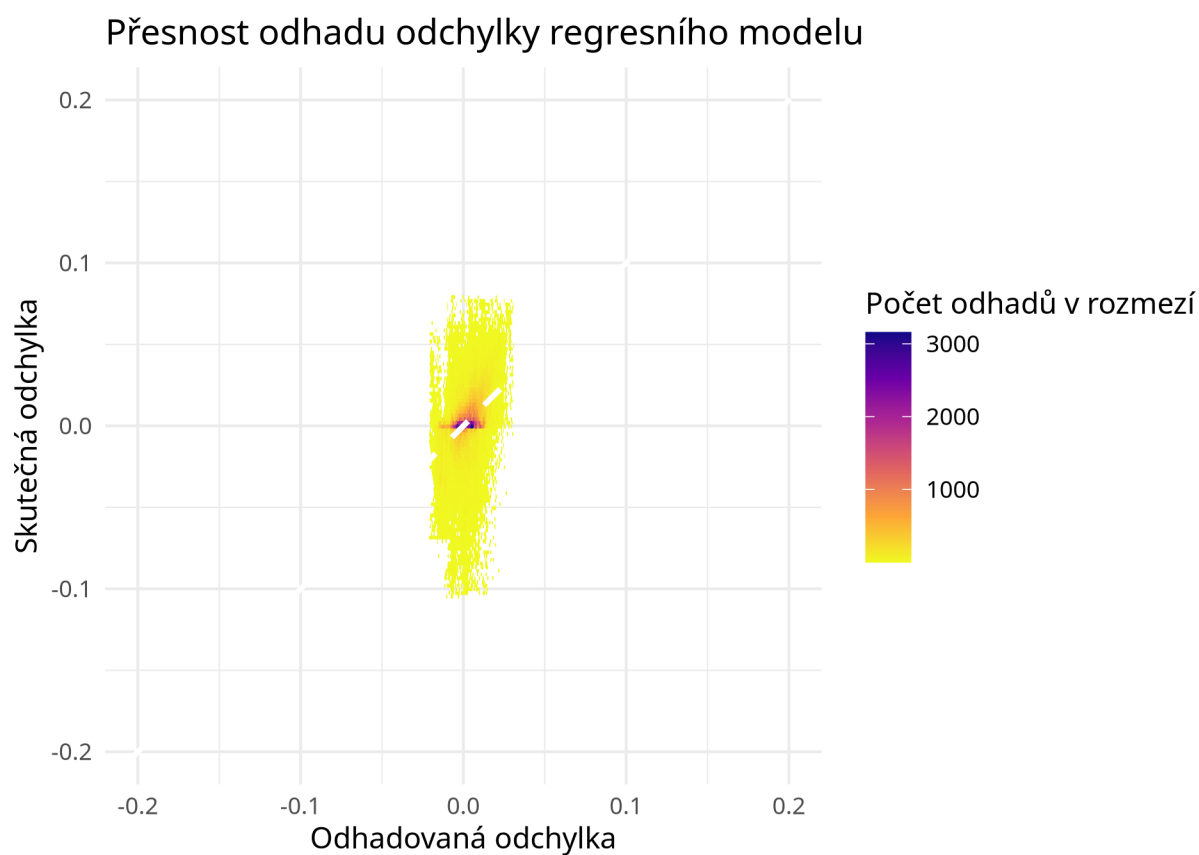
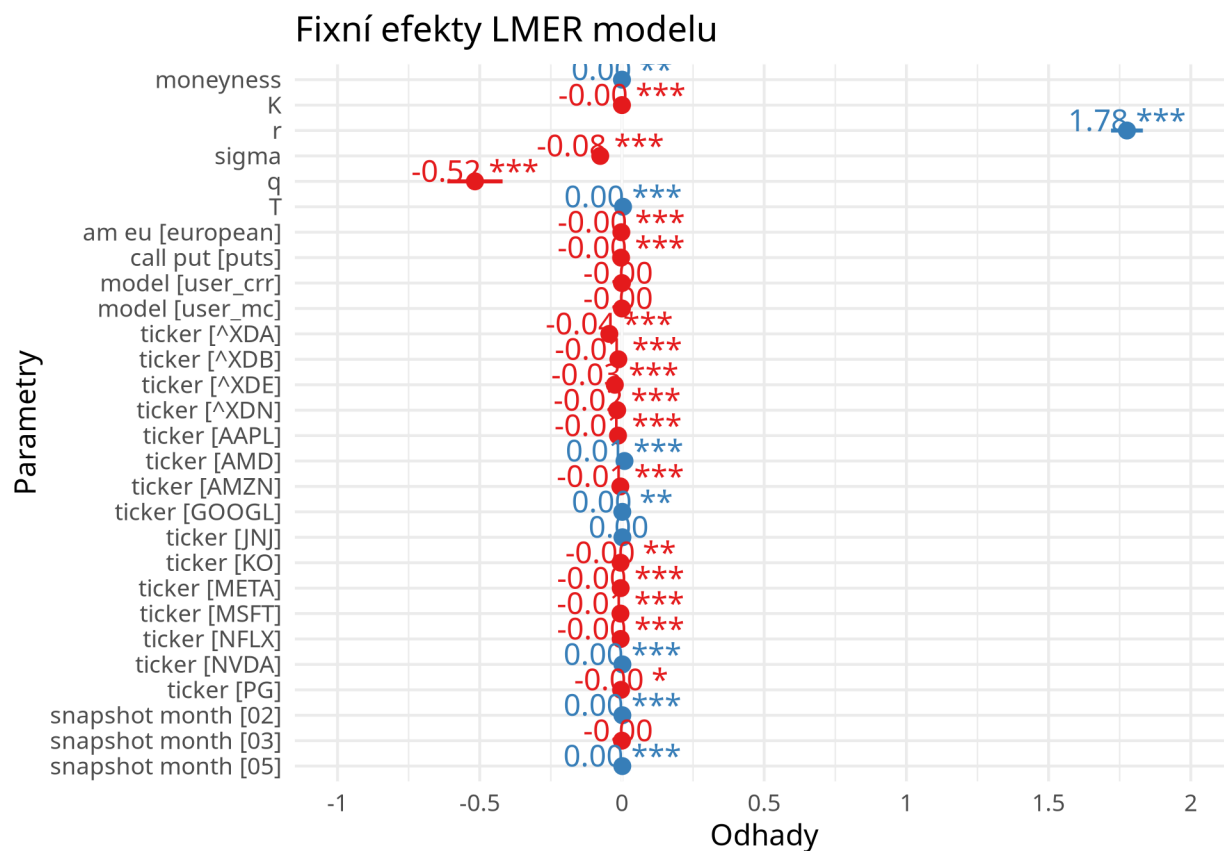
##
## Call:
## lm(formula = formula, data = train_data_rel)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.116244 -0.005499  0.001000  0.007341  0.085975
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -2.943e-02  1.279e-03 -23.013 < 2e-16 ***
## moneyness      9.768e-05  3.310e-05   2.951 0.003168 **
## K             -6.876e-06  1.599e-07 -42.999 < 2e-16 ***
## r              1.776e+00  2.854e-02  62.203 < 2e-16 ***
## sigma        -7.635e-02  1.244e-03 -61.358 < 2e-16 ***
## q             -5.167e-01  4.951e-02 -10.436 < 2e-16 ***
## T              3.834e-03  4.181e-05  91.718 < 2e-16 ***
## am_eueuropean -2.130e-03  5.683e-04  -3.748 0.000178 ***
## call_putputs  -3.068e-03  6.041e-05 -50.779 < 2e-16 ***
## modeluser_crr -1.476e-05  7.081e-05  -0.208 0.834903
## modeluser_mc  -5.503e-05  7.085e-05  -0.777 0.437304
```

```
## ticker^XDA      -4.432e-02  1.929e-03 -22.980 < 2e-16 ***
## ticker^XDB      -1.256e-02  1.625e-03  -7.731 1.07e-14 ***
## ticker^XDE      -2.507e-02  1.180e-03 -21.248 < 2e-16 ***
## ticker^XDN      -1.685e-02  1.431e-03 -11.776 < 2e-16 ***
## tickerAAPL      -1.387e-02  4.239e-04 -32.719 < 2e-16 ***
## tickerAMD        8.940e-03  1.571e-04  56.899 < 2e-16 ***
## tickerAMZN      -5.636e-03  3.181e-04 -17.719 < 2e-16 ***
## tickerGOOGL      1.245e-03  4.079e-04   3.053 0.002266 **
## tickerJNJ        1.671e-03  1.738e-03   0.961 0.336381
## tickerKO        -4.758e-03  1.587e-03  -2.998 0.002719 **
## tickerMETA      -4.490e-03  3.017e-04 -14.880 < 2e-16 ***
## tickerMSFT      -5.288e-03  5.642e-04  -9.372 < 2e-16 ***
## tickerNFLX      -4.631e-03  3.123e-04 -14.831 < 2e-16 ***
## tickerNVDA       1.677e-03  2.265e-04   7.404 1.32e-13 ***
## tickerPG        -3.174e-03  1.439e-03  -2.205 0.027431 *
## tickerTSLA       NA          NA          NA          NA
## snapshot_month02  1.472e-03  8.430e-05  17.465 < 2e-16 ***
## snapshot_month03 -2.282e-05  8.941e-05  -0.255 0.798513
## snapshot_month05  1.251e-03  1.091e-04  11.460 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0166 on 329483 degrees of freedom
## Multiple R-squared:  0.1728, Adjusted R-squared:  0.1728
## F-statistic: 2459 on 28 and 329483 DF, p-value: < 2.2e-16
```

```
r2(rmr_relative)
```

```
## # R2 for Linear Regression
##      R2: 0.173
##   adj. R2: 0.173

## Model matrix is rank deficient. Parameters `tickerTSLA` were not
## estimable.
```



```
anova(rmr_relative)
```

```
## Analysis of Variance Table
##
## Response: pricing_error
##           Df Sum Sq Mean Sq    F value Pr(>F)
## moneyness    1  0.124   0.1242    450.5876 <2e-16 ***
## K             1  0.690   0.6898   2503.0910 <2e-16 ***
## r            1  1.935   1.9346   7020.1455 <2e-16 ***
## sigma        1  6.870   6.8704  24930.8467 <2e-16 ***
## q            1  1.141   1.1413   4141.6189 <2e-16 ***
## T            1  2.256   2.2561   8186.8529 <2e-16 ***
## am_eu        1  0.110   0.1102    400.0206 <2e-16 ***
## call_put     1  0.772   0.7721   2801.6368 <2e-16 ***
## model        2  0.000   0.0001     0.2836  0.753
## ticker      15  4.938   0.3292   1194.5631 <2e-16 ***
## snapshot_month 3  0.136   0.0453    164.5608 <2e-16 ***
## Residuals   329483 90.798   0.0003
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ranef(rmr_relative)
```

LMER model - náhodné efekty, K

Mnohem lepší R^2 , stále malé. Nevýznamný poměr S/K a styl využití opce (a model).

```
subset_args <- c(
  #"S",
  "moneyness",
  "K",
  "r",
  "sigma",
  "q",
  "T",
  "am_eu",
  "call_put",
  "model"
  #"ticker"
)
interactions <- c(
  "(1 | ticker)",
  "(1 | snapshot_month)"
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lmer(formula, data = train_data_rel)
```

```
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
```

```
summary(rmr_relative)
```

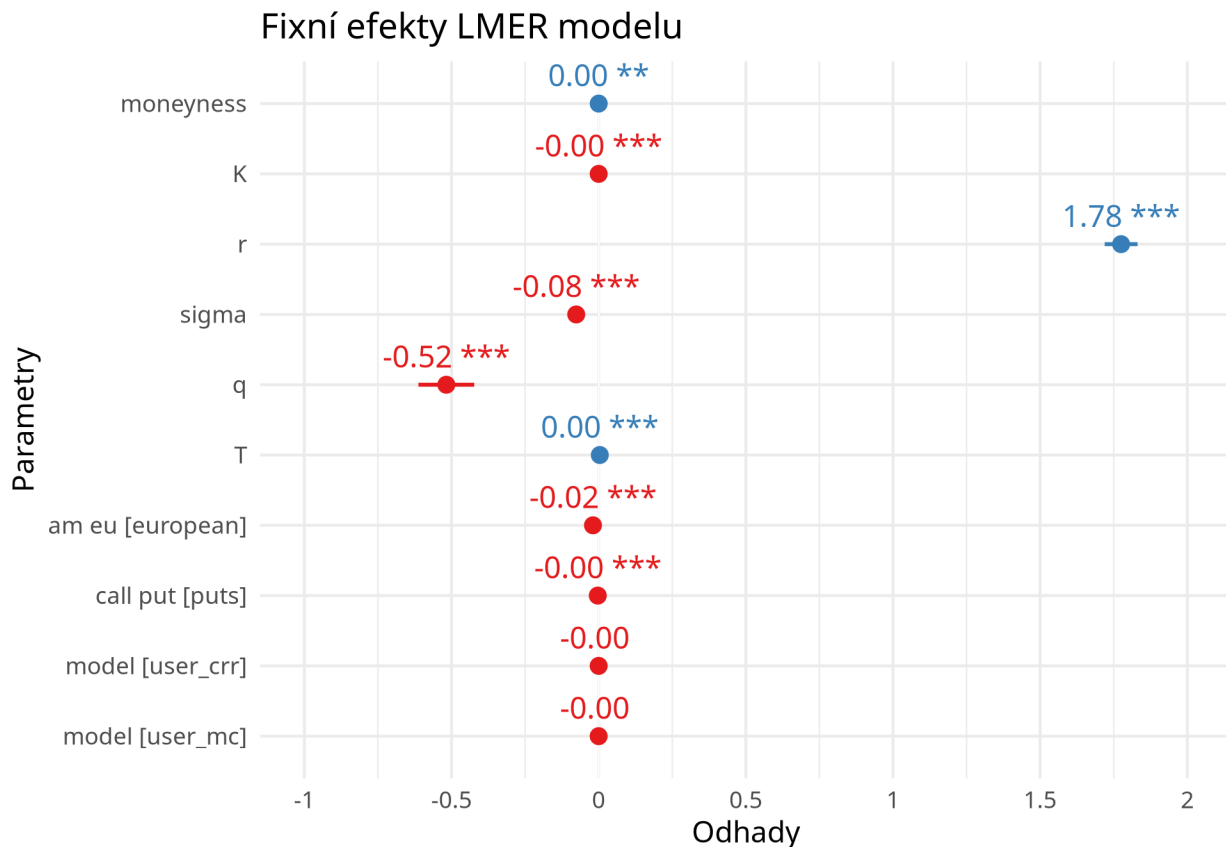
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: formula
## Data: train_data_rel
##
## REML criterion at convergence: -1765482
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.0018 -0.3312  0.0603  0.4421  5.1790
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## ticker          (Intercept) 9.274e-05 0.0096303
## snapshot_month (Intercept) 6.348e-07 0.0007967
## Residual                2.756e-04 0.0166005
## Number of obs: 329512, groups: ticker, 17; snapshot_month, 4
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -3.112e-02  3.066e-03  2.132e+01 -10.148 1.28e-09 ***
## moneyness     9.925e-05  3.310e-05  3.284e+05  2.999  0.00271 **
## K             -6.861e-06  1.598e-07  3.163e+05 -42.939 < 2e-16 ***
## r             1.775e+00  2.850e-02  1.758e+05  62.275 < 2e-16 ***
## sigma        -7.630e-02  1.240e-03  1.024e+05 -61.544 < 2e-16 ***
## q            -5.176e-01  4.834e-02  6.502e+03 -10.709 < 2e-16 ***
## T             3.834e-03  4.181e-05  3.294e+05  91.704 < 2e-16 ***
## am_eueuropean -1.939e-02  5.190e-03  1.520e+01  -3.735  0.00195 **
## call_putputs  -3.066e-03  6.041e-05  3.295e+05 -50.752 < 2e-16 ***
## modelusr_crr  -1.471e-05  7.081e-05  3.295e+05  -0.208  0.83538
## modelusr_mc   -5.500e-05  7.085e-05  3.295e+05  -0.776  0.43757
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) mnyss K      r      sigma q      T      am_rpn cll_pt
## moneyness    -0.020
## K            -0.017  0.501
## r            -0.343  0.020  0.008
## sigma        -0.142 -0.058 -0.069  0.001
## q            -0.147  0.010  0.004 -0.025  0.048
## T            -0.059 -0.101 -0.075  0.157 -0.007  0.000
## am_eueuropn  -0.506 -0.006 -0.012 -0.002  0.059  0.090  0.003
## call_putpts  -0.006 -0.037  0.211 -0.010 -0.013 -0.001 -0.019 -0.002
## modelsr_crr  -0.011 -0.001  0.000 -0.001  0.000 -0.001  0.000  0.000  0.000
## modelusr_mc  -0.011 -0.002 -0.001 -0.001  0.000  0.000  0.000  0.000  0.000
##              mdlr_c
## moneyness
## K
## r
## sigma
## q
```

```
## T
## am_eueuropn
## call_putpts
## modelsr_crr
## modelusr_mc 0.500
## fit warnings:
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
r2(rmr_relative)

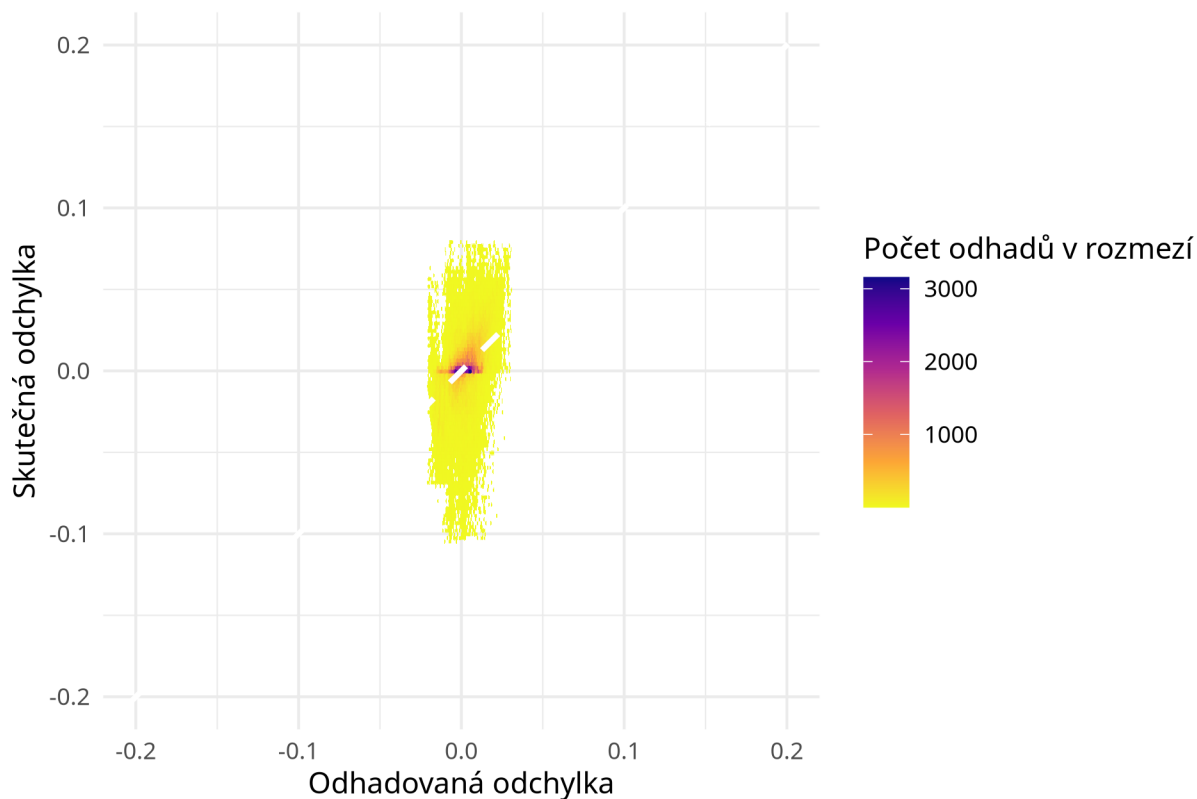
## Warning: Can't compute random effect variances. Some variance components equal
## zero. Your model may suffer from singularity (see `?lme4::isSingular`
## and `?performance::check_singularity`).
## Decrease the `tolerance` level to force the calculation of random effect
## variances, or impose priors on your random effects parameters (using
## packages like `brms` or `glmmTMB`).

## Random effect variances not available. Returned R2 does not account for random effects.

## # R2 for Mixed Models
##
## Conditional R2: NA
## Marginal R2: 0.319
```



Přesnost odhadu odchylny regresního modelu



```
anova(rmr_relative)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq Mean Sq NumDF  DenDF    F value    Pr(>F)
## moneyness 0.00248 0.00248     1 328370    8.9934 0.002710 **
## K          0.50811 0.50811     1 316310 1843.7988 < 2.2e-16 ***
## r          1.06873 1.06873     1 175814 3878.1228 < 2.2e-16 ***
## sigma      1.04381 1.04381     1 102445 3787.7217 < 2.2e-16 ***
## q          0.03160 0.03160     1   6502  114.6817 < 2.2e-16 ***
## T          2.31753 2.31753     1 329446 8409.6917 < 2.2e-16 ***
## am_eu      0.00384 0.00384     1    15    13.9484 0.001952 **
## call_put   0.70984 0.70984     1 329481 2575.8146 < 2.2e-16 ***
## model      0.00018 0.00009     2 329483    0.3229 0.724042
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ranef(rmr_relative)
```

```
## $ticker
##      (Intercept)
## ^RUT  0.0195770209
## ^XDA -0.0237926181
## ^XDB  0.0068772912
## ^XDE -0.0053787670
## ^XDN  0.0027170730
## AAPL -0.0115115258
## AMD   0.0112737067
## AMZN -0.0032837207
```

```
## GOOGL 0.0036015244
## JNJ 0.0040604829
## KO -0.0023659079
## META -0.0021435765
## MSFT -0.0029264114
## NFLX -0.0022776730
## NVDA 0.0040244540
## PG -0.0007873576
## TSLA 0.0023360061
##
## $snapshot_month
## (Intercept)
## 01 -0.0006723152
## 02 0.0007914657
## 03 -0.0006936764
## 05 0.0005745259
##
## with conditional variances for "ticker" "snapshot_month"
```

LMER model - fixní efekty, S

Mnohem lepší R^2 , stále malé. Nevýznamný poměr S/K a styl využití opce (a model).

```
subset_args <- c(
  "S",
  "moneyness",
  #"K",
  "r",
  "sigma",
  "q",
  "T",
  "am_eu",
  "call_put",
  "model",
  "ticker",
  "snapshot_month"
)
interactions <- c()

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lm(formula, data = train_data_rel)

summary(rmr_relative)
```

```
##
## Call:
## lm(formula = formula, data = train_data_rel)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.116219 -0.005524  0.000993  0.007449  0.083786
##
## Coefficients: (1 not defined because of singularities)
```

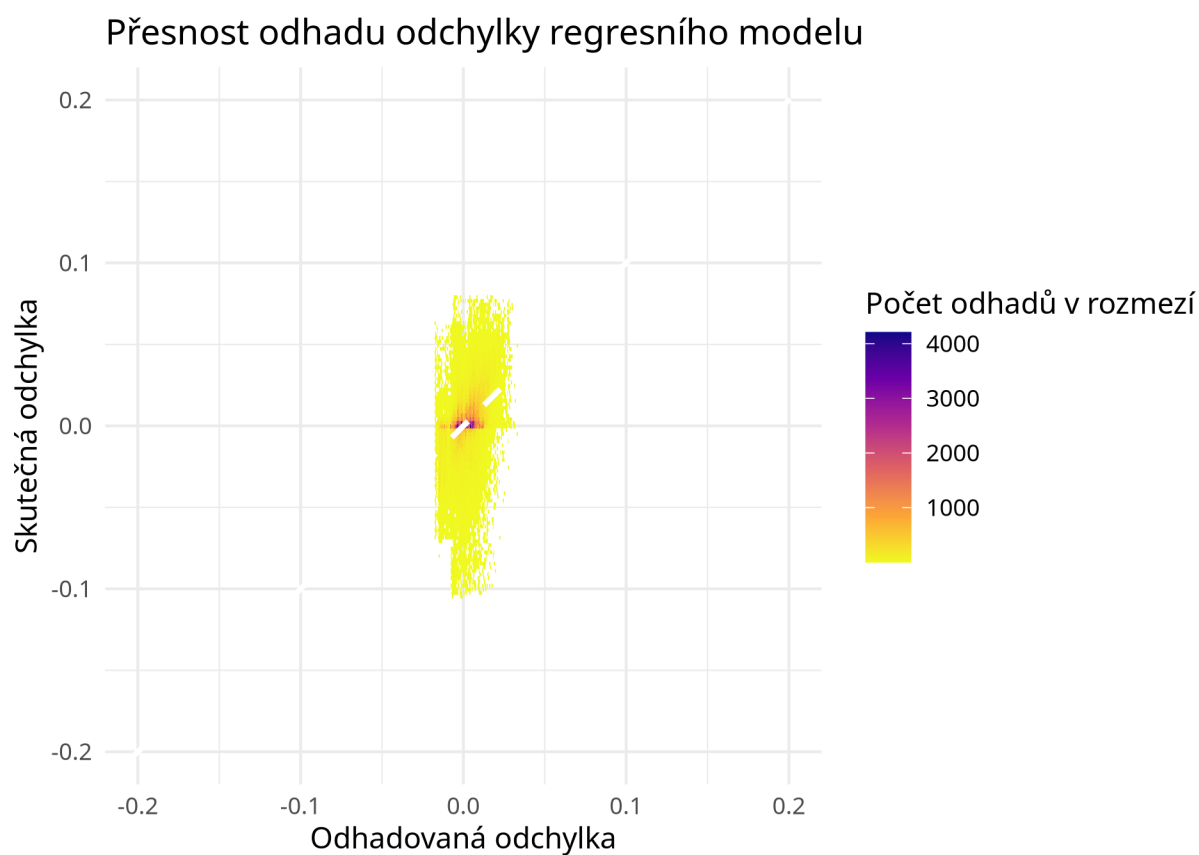
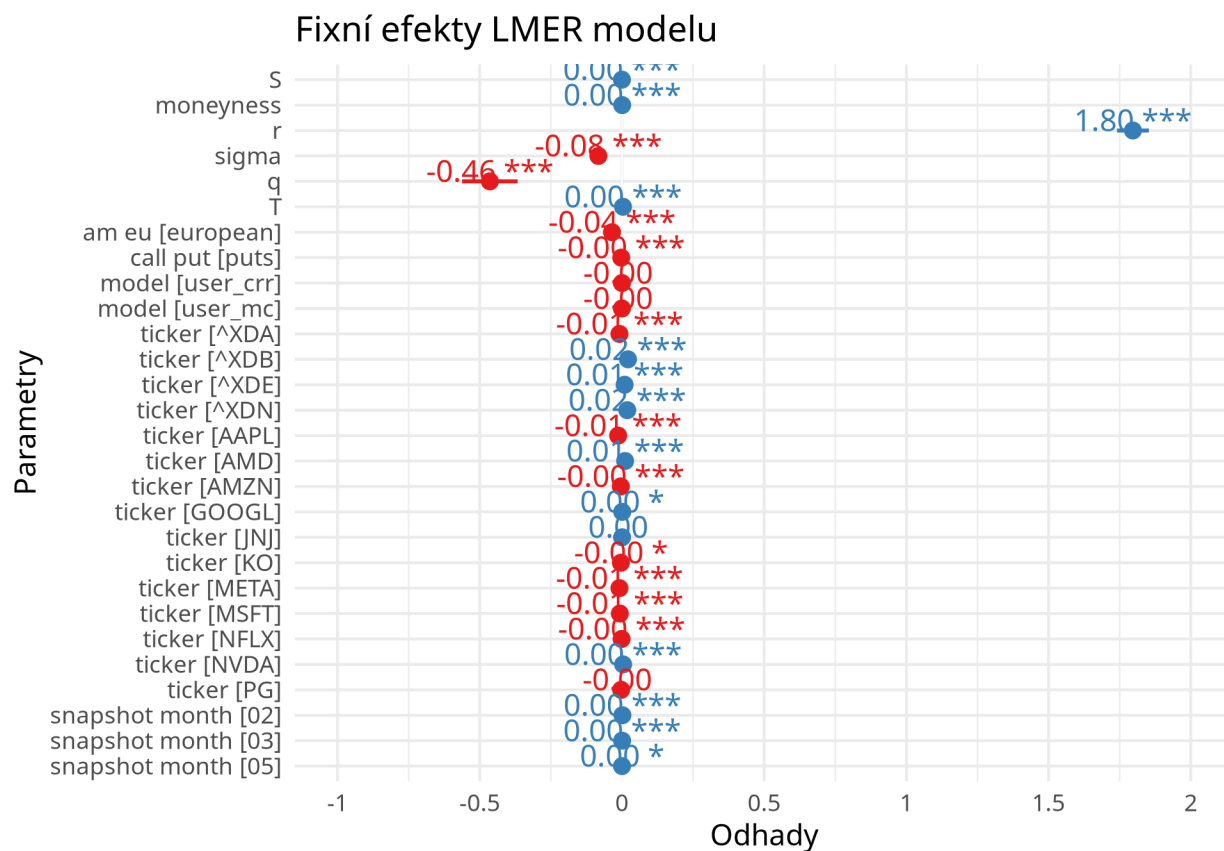
```

##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -3.375e-02  1.290e-03 -26.161 < 2e-16 ***
## S              7.251e-06  5.999e-07  12.087 < 2e-16 ***
## moneyness      8.024e-04  2.873e-05  27.927 < 2e-16 ***
## r              1.796e+00  2.863e-02  62.746 < 2e-16 ***
## sigma         -8.284e-02  1.266e-03 -65.440 < 2e-16 ***
## q             -4.644e-01  4.977e-02  -9.330 < 2e-16 ***
## T              3.702e-03  4.180e-05  88.559 < 2e-16 ***
## am_european   -3.502e-02  1.506e-03 -23.261 < 2e-16 ***
## call_puts     -2.514e-03  5.920e-05 -42.465 < 2e-16 ***
## modeluser_crr -1.585e-05  7.099e-05  -0.223 0.823332
## modeluser_mc  -5.610e-05  7.103e-05  -0.790 0.429661
## ticker^XDA    -8.834e-03  2.439e-03  -3.621 0.000293 ***
## ticker^XDB     2.129e-02  2.179e-03   9.772 < 2e-16 ***
## ticker^XDE     9.336e-03  1.876e-03   4.977 6.45e-07 ***
## ticker^XDN     1.870e-02  2.072e-03   9.027 < 2e-16 ***
## tickerAAPL    -1.345e-02  4.250e-04 -31.645 < 2e-16 ***
## tickerAMD      1.119e-02  1.755e-04  63.794 < 2e-16 ***
## tickerAMZN    -4.030e-03  3.235e-04 -12.460 < 2e-16 ***
## tickerGOOGL    9.788e-04  4.096e-04   2.390 0.016866 *
## tickerJNJ      5.074e-04  1.746e-03   0.291 0.771325
## tickerKO      -3.745e-03  1.591e-03  -2.354 0.018573 *
## tickerMETA    -8.830e-03  3.468e-04 -25.460 < 2e-16 ***
## tickerMSFT    -7.426e-03  5.742e-04 -12.933 < 2e-16 ***
## tickerNFLX    -1.248e-03  3.414e-04  -3.654 0.000258 ***
## tickerNVDA     4.047e-03  2.465e-04  16.419 < 2e-16 ***
## tickerPG      -2.826e-03  1.443e-03  -1.958 0.050172 .
## tickerTSLA      NA         NA         NA         NA
## snapshot_month02 1.640e-03  8.513e-05  19.261 < 2e-16 ***
## snapshot_month03 5.114e-04  9.636e-05   5.307 1.12e-07 ***
## snapshot_month05 2.672e-04  1.154e-04   2.317 0.020528 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01664 on 329483 degrees of freedom
## Multiple R-squared:  0.1686, Adjusted R-squared:  0.1685
## F-statistic: 2386 on 28 and 329483 DF, p-value: < 2.2e-16
r2(rmr_relative)

## # R2 for Linear Regression
##      R2: 0.169
##   adj. R2: 0.168

## Model matrix is rank deficient. Parameters `tickerTSLA` were not
## estimable.

```



```
anova(rmr_relative)
```

```
## Analysis of Variance Table
##
## Response: pricing_error
##          Df Sum Sq Mean Sq    F value Pr(>F)
## S          1  0.423   0.4234  1528.5356 <2e-16 ***
## moneyness   1  0.093   0.0932   336.5894 <2e-16 ***
## r           1  1.893   1.8934  6835.1815 <2e-16 ***
## sigma       1  6.859   6.8586 24760.0831 <2e-16 ***
## q           1  1.096   1.0960  3956.4833 <2e-16 ***
## T           1  2.149   2.1494  7759.4271 <2e-16 ***
## am_eu       1  0.069   0.0693   250.1558 <2e-16 ***
## call_put    1  0.520   0.5199  1876.9180 <2e-16 ***
## model       2  0.000   0.0001    0.3163 0.7288
## ticker     15  5.295   0.3530  1274.2803 <2e-16 ***
## snapshot_month 3  0.106   0.0353   127.3656 <2e-16 ***
## Residuals 329483 91.267  0.0003
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ranef(rmr_relative)
```

LMER model - náhodné efekty, S

Mnohem lepší R^2 , stále malé. Nevýznamný poměr S/K a styl využití opce (a model).

```
subset_args <- c(
  "S",
  "moneyness",
  #"K",
  "r",
  "sigma",
  "q",
  "T",
  "am_eu",
  "call_put",
  "model"
  #"ticker"
)
interactions <- c(
  "(1 | ticker)",
  "(1 | snapshot_month)"
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lmer(formula, data = train_data_rel)
```

```
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
```

```
summary(rmr_relative)
```

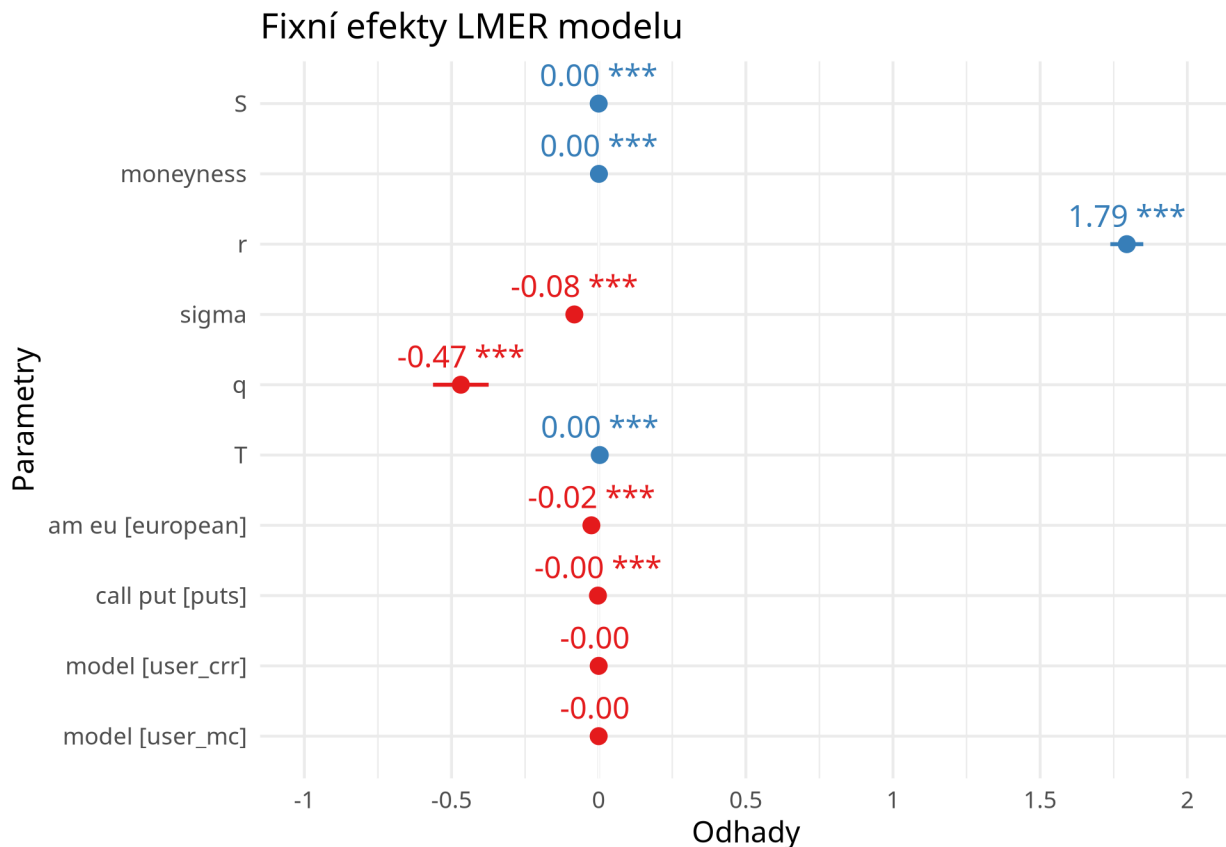
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: formula
## Data: train_data_rel
##
## REML criterion at convergence: -1763792
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.9820 -0.3319  0.0596  0.4473  5.0350
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## ticker           (Intercept) 6.971e-05 0.0083493
## snapshot_month (Intercept) 5.123e-07 0.0007158
## Residual                2.770e-04 0.0166434
## Number of obs: 329512, groups: ticker, 17; snapshot_month, 4
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -3.512e-02  2.734e-03 2.350e+01 -12.847 4.09e-12 ***
## S             7.104e-06  5.902e-07 1.612e+04  12.036 < 2e-16 ***
## moneyness    8.024e-04  2.873e-05 3.295e+05  27.927 < 2e-16 ***
## r            1.794e+00  2.858e-02 1.374e+05  62.792 < 2e-16 ***
## sigma       -8.264e-02  1.259e-03 5.332e+04 -65.645 < 2e-16 ***
## q           -4.684e-01  4.821e-02 3.707e+03  -9.716 < 2e-16 ***
## T            3.701e-03  4.180e-05 3.294e+05  88.551 < 2e-16 ***
## am_eueuropean -2.478e-02  4.523e-03 1.531e+01  -5.479 5.90e-05 ***
## call_putputs -2.513e-03  5.920e-05 3.295e+05 -42.457 < 2e-16 ***
## modelusr_crr -1.580e-05  7.099e-05 3.295e+05  -0.223  0.824
## modelusr_mc  -5.606e-05  7.103e-05 3.295e+05  -0.789  0.430
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) S      mnyyss r      sigma  q      T      am_rpn cll_pt
## S              -0.054
## moneyness     -0.014 -0.024
## r              -0.387  0.029  0.017
## sigma         -0.149 -0.179 -0.023 -0.003
## q             -0.168  0.071  0.008 -0.022  0.038
## T             -0.069  0.003 -0.074  0.158 -0.013  0.001
## am_eueuropn   -0.494 -0.050  0.002 -0.004  0.075  0.100  0.002
## call_putpts   -0.003  0.007 -0.169 -0.012  0.000 -0.002 -0.003  0.000
## modelsr_crr   -0.013  0.000  0.000 -0.001  0.000 -0.001  0.000  0.000  0.000
## modelusr_mc   -0.013  0.001 -0.001 -0.001  0.000  0.000  0.000  0.000  0.000
##              mdlusr_c
## S
## moneyness
## r
## sigma
## q
```

```
## T
## am_eueuropn
## call_putpts
## modelsr_crr
## modelusr_mc 0.500
## fit warnings:
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
r2(rmr_relative)

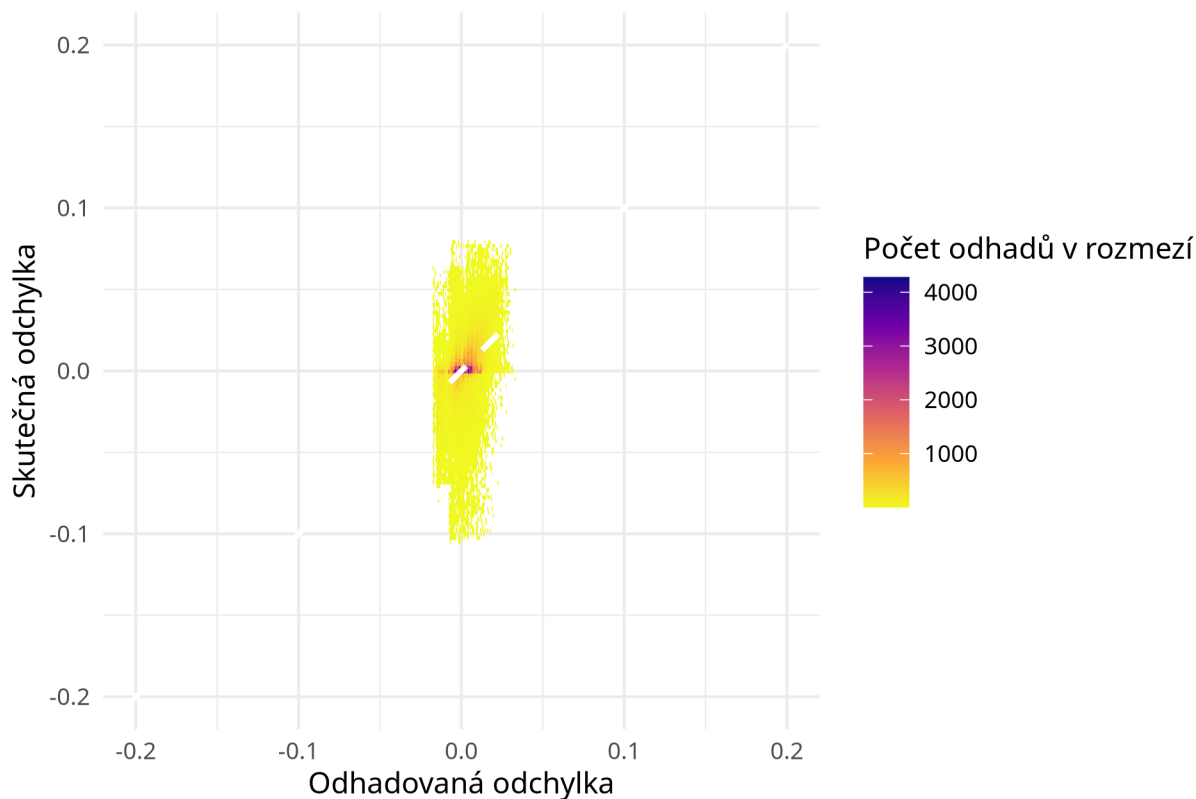
## Warning: Can't compute random effect variances. Some variance components equal
## zero. Your model may suffer from singularity (see `?lme4::isSingular`
## and `?performance::check_singularity`).
## Decrease the `tolerance` level to force the calculation of random effect
## variances, or impose priors on your random effects parameters (using
## packages like `brms` or `glmmTMB`).

## Random effect variances not available. Returned R2 does not account for random effects.

## # R2 for Mixed Models
##
## Conditional R2: NA
## Marginal R2: 0.246
```



Přesnost odhadu odchylny regresního modelu



```
anova(rmr_relative)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##          Sum Sq Mean Sq NumDF  DenDF  F value    Pr(>F)
## S          0.04013  0.04013      1   16123   144.867 < 2.2e-16 ***
## moneyness  0.21604  0.21604      1  329498   779.910 < 2.2e-16 ***
## r          1.09217  1.09217      1 137377   3942.815 < 2.2e-16 ***
## sigma      1.19369  1.19369      1   53316  4309.307 < 2.2e-16 ***
## q          0.02615  0.02615      1    3707    94.409 < 2.2e-16 ***
## T          2.17204  2.17204      1  329397  7841.243 < 2.2e-16 ***
## am_eu      0.00832  0.00832      1     15    30.019 5.903e-05 ***
## call_put   0.49933  0.49933      1  329497  1802.610 < 2.2e-16 ***
## model      0.00018  0.00009      2  329483     0.331   0.7182
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ranef(rmr_relative)
```

```
## $ticker
##      (Intercept)
## ^RUT  -0.0078312635
## ^XDA  -0.0162007009
## ^XDB   0.0126606687
## ^XDE   0.0011352850
## ^XDN   0.0102360106
## AAPL  -0.0113890886
## AMD    0.0131711571
## AMZN  -0.0020050643
```

```
## GOOGL  0.0030418080
## JNJ    0.0026957701
## KO     -0.0015875867
## META   -0.0067428299
## MSFT   -0.0053244298
## NFLX    0.0007565412
## NVDA    0.0060507186
## PG     -0.0006749605
## TSLA    0.0020079642
##
## $snapshot_month
##      (Intercept)
## 01 -0.0006025341
## 02  0.0010221480
## 03 -0.0001004766
## 05 -0.0003191372
##
## with conditional variances for "ticker" "snapshot_month"
```